

Universidad Central del Ecuador
Facultad de Filosofía Letras y Ciencias de la Educación

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Curso: 5to "B"

Fecha:

Vectores (3D)

1) $\vec{A} = 2\hat{i} + 3\hat{j} + 4\hat{k} \text{ (m)}$
 $\vec{B} = 1\hat{i} + 2\hat{j} + 5\hat{k} \text{ (m)}$

$A_x = 2, A_y = 3, A_z = 4 \text{ (m)}$

$B_x = 1, B_y = 2, B_z = 5 \text{ (m)}$

$\vec{R} = \vec{A} + \vec{B}$

$R_x = A_x + B_x = 2 + 1 = 3 \text{ (m)}$

$R_y = A_y + B_y = 3 + 2 = 5 \text{ (m)}$

$R_z = A_z + B_z = 4 + 5 = 9 \text{ (m)}$

$\vec{R} = 3\hat{i} + 5\hat{j} + 9\hat{k} \text{ (m)}$

$R = \sqrt{R_x^2 + R_y^2 + R_z^2}$

$R = \sqrt{3^2 + 5^2 + 9^2}$

$R = \sqrt{9 + 25 + 81}$

$R = \sqrt{115}$

$R = 10.72 \text{ (m)}$

$\alpha = \cos^{-1}\left(\frac{R_x}{R}\right)$

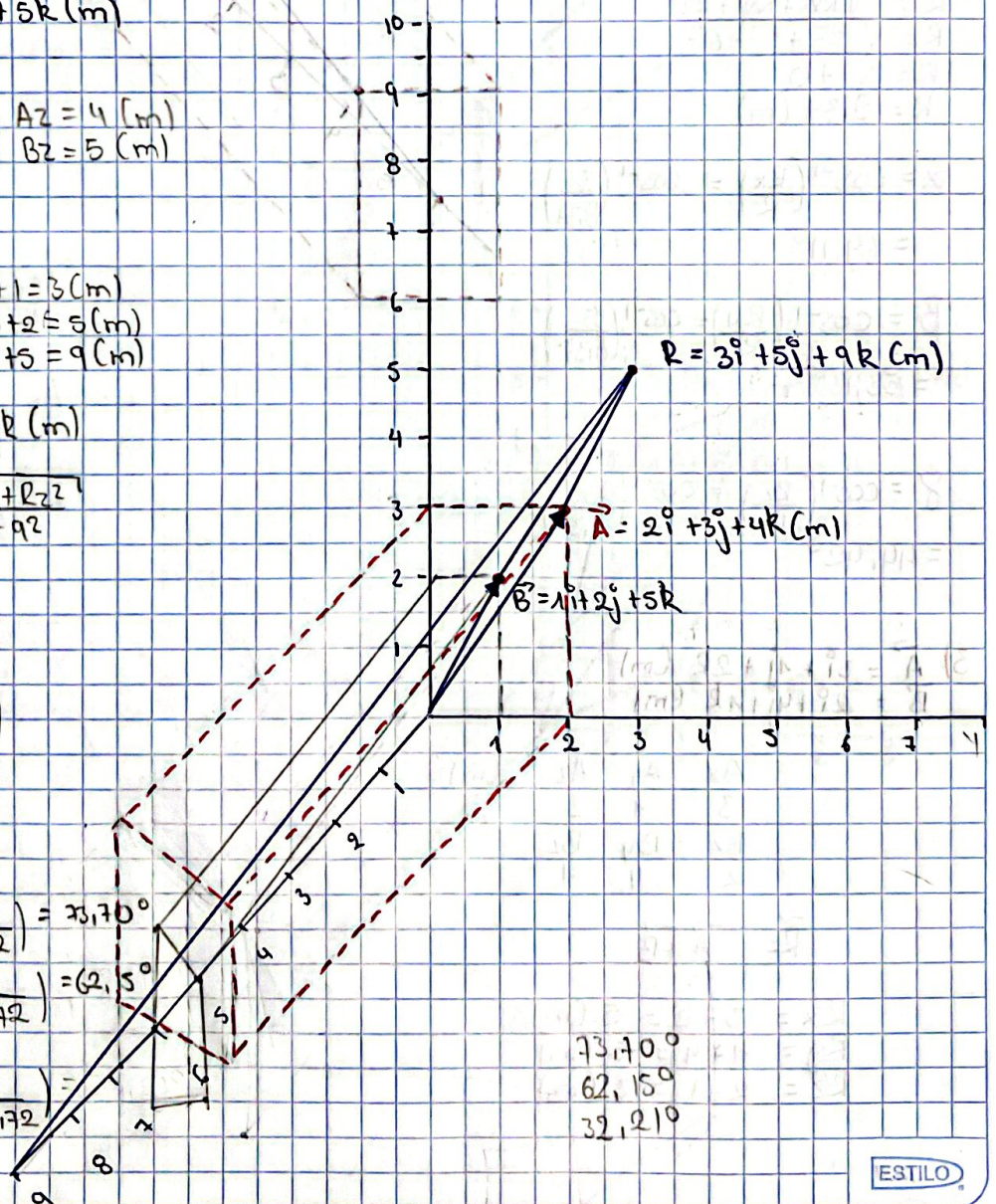
$\beta = \cos^{-1}\left(\frac{R_y}{R}\right)$

$\gamma = \cos^{-1}\left(\frac{R_z}{R}\right)$

$\alpha = \cos^{-1}\left(\frac{3}{10.72}\right) = 73.70^\circ$

$\beta = \cos^{-1}\left(\frac{5}{10.72}\right) = 62.15^\circ$

$\gamma = \cos^{-1}\left(\frac{9}{10.72}\right) = 32.21^\circ$



ESTILO

2) $\vec{A} = 2\hat{i} + 3\hat{j} + 4\hat{k} \text{ (m)}$
 $\vec{B} = 1\hat{i} + 2\hat{j} + 2\hat{k} \text{ (m)}$

$A_x \quad A_y \quad A_z \text{ (m)}$

2

3

4

$B_x \quad B_y \quad B_z \text{ (m)}$

1

2

2

$\vec{R} = \vec{A} + \vec{B}$

$R_x = A_x + B_x = 2 + 1 = 3 \text{ (m)}$

$R_y = A_y + B_y = 3 + 2 = 5 \text{ (m)}$

$R_z = A_z + B_z = 4 + 2 = 6 \text{ (m)}$

$\vec{R} = 3\hat{i} + 5\hat{j} + 6\hat{k} \text{ (m)}$

$R = \sqrt{R_x^2 + R_y^2 + R_z^2}$

$R = \sqrt{3^2 + 5^2 + 6^2}$

$R = \sqrt{70}$

$R = 8.37 \text{ (m)}$

$\alpha = \cos^{-1}\left(\frac{R_x}{R}\right) = \cos^{-1}\left(\frac{3}{8.37}\right)$

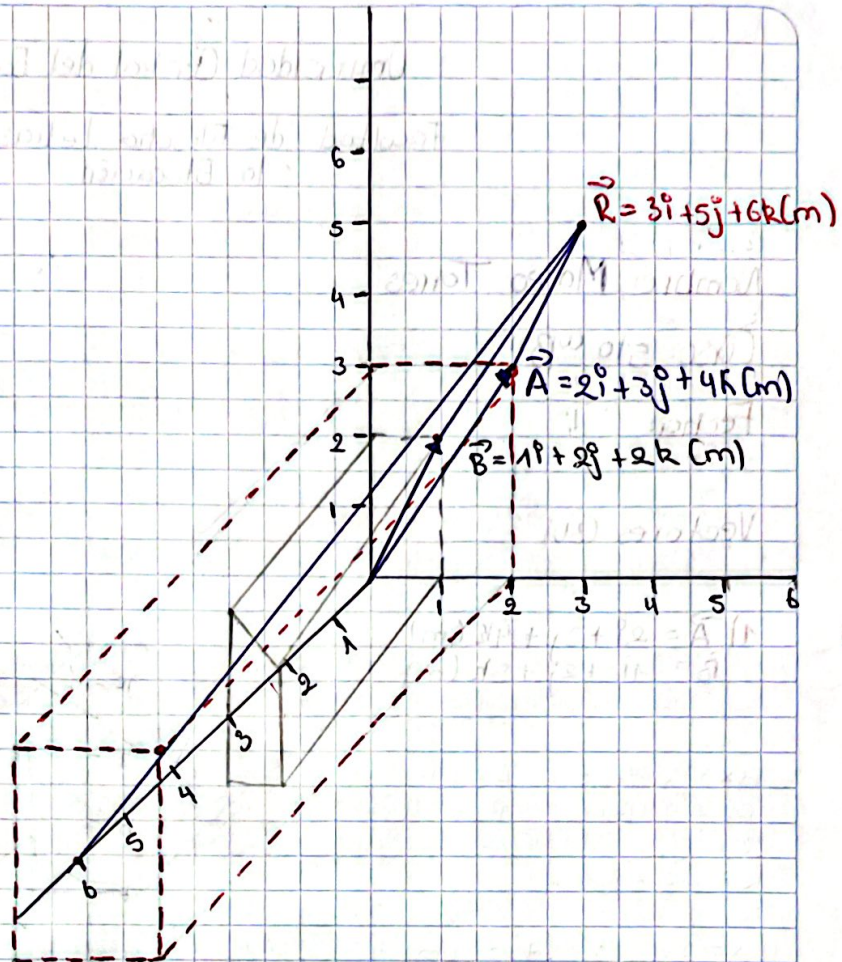
$= 69.11^\circ$

$\beta = \cos^{-1}\left(\frac{R_y}{R}\right) = \cos^{-1}\left(\frac{5}{8.37}\right)$

$= 53.13^\circ$

$\gamma = \cos^{-1}\left(\frac{R_z}{R}\right) = \cos^{-1}\left(\frac{6}{8.37}\right)$

$= 44.42^\circ$



3) $\vec{A} = 3\hat{i} + 1\hat{j} + 2\hat{k} \text{ (m)}$
 $\vec{B} = 2\hat{i} + 4\hat{j} + 1\hat{k} \text{ (m)}$

$A_x \quad A_y \quad A_z \text{ (m)}$

3

1

2

$B_x \quad B_y \quad B_z$

2

4

1

$\vec{R} = \vec{A} + \vec{B}$

$R_x = 3 + 2 = 5 \text{ (m)}$

$R_y = 1 + 4 = 5 \text{ (m)}$

$R_z = 2 + 1 = 3 \text{ (m)}$

$$\vec{R} = 5\mathbf{i} + 5\mathbf{j} + 3\mathbf{k} \text{ (m)}$$

$$R = \sqrt{5^2 + 5^2 + 3^2}$$

$$R = \sqrt{25 + 25 + 9}$$

$$R = \sqrt{59}$$

$$R = 7.68 \text{ (m)}$$

$$\alpha = \cos^{-1}\left(\frac{R_x}{R}\right) = \cos^{-1}\left(\frac{5}{7.68}\right)$$

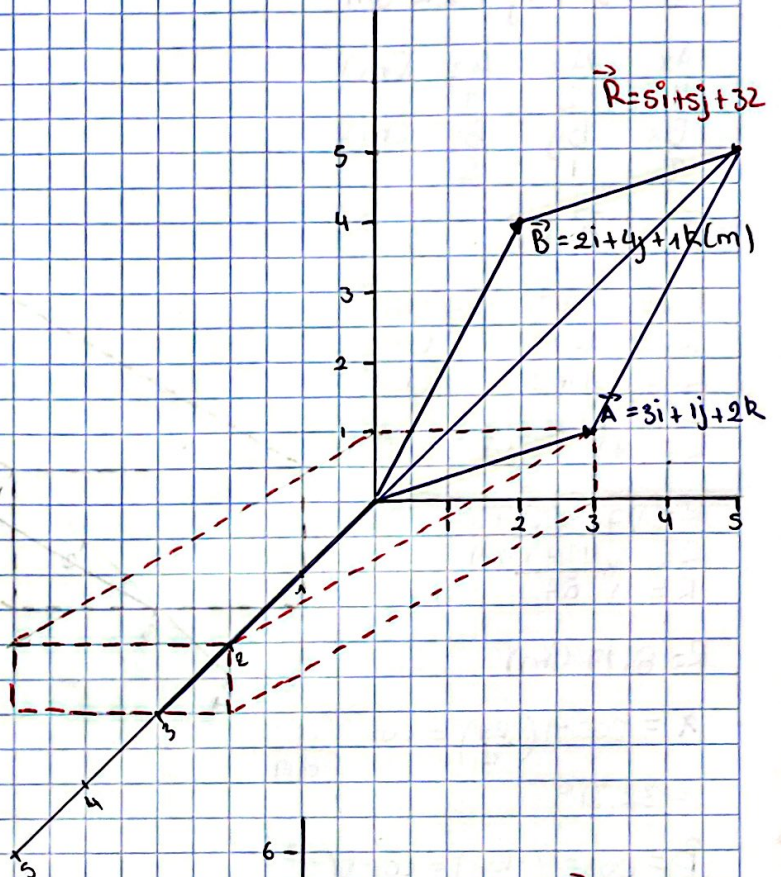
$$= 49.46^\circ$$

$$\beta = \cos^{-1}\left(\frac{R_y}{R}\right) = \cos^{-1}\left(\frac{5}{7.68}\right)$$

$$= 49.46^\circ$$

$$\gamma = \cos^{-1}\left(\frac{R_z}{R}\right) = \cos^{-1}\left(\frac{3}{7.68}\right)$$

$$= 67.79^\circ$$



$$4) \vec{A} = 1\mathbf{i} + 4\mathbf{j} + 3\mathbf{k} \text{ (m)}$$

$$\vec{B} = 2\mathbf{i} + 1\mathbf{j} + 4\mathbf{k} \text{ (m)}$$

$$A_x \quad A_y \quad A_z \quad \text{(m)}$$

$$1 \quad 4 \quad 3$$

$$B_x \quad B_y \quad B_z \quad \text{(m)}$$

$$2 \quad 1 \quad 4$$

$$\vec{R} = \vec{A} + \vec{B}$$

$$R_x = 1 + 2 = 3 \text{ (m)}$$

$$R_y = 4 + 1 = 5 \text{ (m)}$$

$$R_z = 3 + 4 = 7 \text{ (m)}$$

$$\vec{R} = 3\mathbf{i} + 5\mathbf{j} + 7\mathbf{k} \text{ (m)}$$

$$R = \sqrt{3^2 + 5^2 + 7^2}$$

$$R = \sqrt{9 + 25 + 49}$$

$$R = \sqrt{83}$$

$$R = 9.11 \text{ (m)}$$

$$\alpha = \cos^{-1}\left(\frac{R_x}{R}\right) = \cos^{-1}\left(\frac{3}{9.11}\right)$$

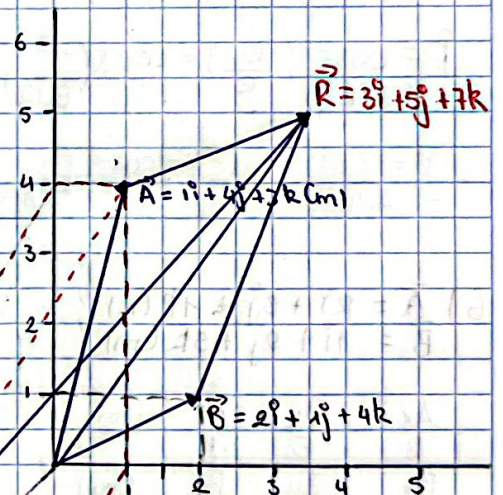
$$= 70.73^\circ$$

$$\beta = \cos^{-1}\left(\frac{R_y}{R}\right) = \cos^{-1}\left(\frac{5}{9.11}\right)$$

$$= 56.77^\circ$$

$$\gamma = \cos^{-1}\left(\frac{R_z}{R}\right) = \cos^{-1}\left(\frac{7}{9.11}\right)$$

$$= 40.02^\circ$$



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$$5) \vec{A} = 4\hat{i} + 2\hat{j} + 1\hat{k} \text{ (m)}$$

$$\vec{B} = 3\hat{i} + 1\hat{j} + 2\hat{k} \text{ (m)}$$

A_x	A_y	A_z	(m)
4	2	1	

B_x	B_y	B_z	(m)
3	1	2	

$$\vec{R} = \vec{A} + \vec{B}$$

$$R_x = 4 + 3 = 7 \text{ (m)}$$

$$R_y = 2 + 1 = 3 \text{ (m)}$$

$$R_z = 1 + 2 = 3 \text{ (m)}$$

$$\vec{R} = 7\hat{i} + 3\hat{j} + 3\hat{k} \text{ (m)}$$

$$R = \sqrt{7^2 + 3^2 + 3^2}$$

$$R = \sqrt{49 + 9 + 9}$$

$$R = \sqrt{67}$$

$$R = 8.19 \text{ (m)}$$

$$\alpha = \cos^{-1}\left(\frac{R_x}{R}\right) = \cos^{-1}\left(\frac{7}{8.19}\right)$$

$$= 32.31^\circ$$

$$\beta = \cos^{-1}\left(\frac{R_y}{R}\right) = \cos^{-1}\left(\frac{3}{8.19}\right)$$

$$= 67.419^\circ$$

$$\gamma = \cos^{-1}\left(\frac{R_z}{R}\right) = \cos^{-1}\left(\frac{3}{8.19}\right)$$

$$= 67.419^\circ$$

$$6) \vec{A} = 2\hat{i} + 5\hat{j} + 1\hat{k} \text{ (m)}$$

$$\vec{B} = 1\hat{i} + 2\hat{j} + 5\hat{k} \text{ (m)}$$

A_x	A_y	A_z	(m)
2	5	1	

B_x	B_y	B_z	(m)
1	2	5	

$$\vec{R} = \vec{A} + \vec{B}$$

$$R_x = 2 + 1 = 3 \text{ (m)}$$

$$R_y = 5 + 2 = 7 \text{ (m)}$$

$$R_z = 1 + 5 = 6 \text{ (m)}$$

$$\vec{R} = 3\hat{i} + 7\hat{j} + 6\hat{k} \text{ (m)}$$

$$R = \sqrt{3^2 + 7^2 + 6^2}$$

$$R = \sqrt{9 + 49 + 36}$$

